

Chemical Kinetics

Outline *As seen in the previous handout, thermodynamic laws allow determination of the equilibrium state of a chemical reaction system. Assuming very fast (instantaneous) reactions as compared to heat and mass transport processes, one can assume local equilibrium conditions and use thermodynamic laws to determine final state. In most combustion cases, however, chemical reactions occur on time scales comparable to that of the flow and molecular transport processes. Therefore, there is a need to determine the chemical reaction rates. In this handout, we will first see the distinction between global and elemental reactions and some usual assumptions in multi-step reactions. Following this, we will formulate reaction rates using different existent theories like collision theory.*

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1 Global and Elementary Reactions

An **elementary** reaction is one that occurs on a molecular level exactly in the way which is described by the reaction equation.



The equation above is an elementary reaction. On the contrary, the following is not an elementary reaction but a **global** or **overall** reaction.:



A general global reaction mechanism involving overall reaction of a moles of oxidizer with one mole of fuel to form b moles of products:



The law of mass action, which is confirmed by numerous experimental observations yield the rate at which fuel is consumed as

$$\frac{d[\text{F}]}{dt} = -k_G(T)[\text{F}]^n[\text{Ox}]^m \quad (4)$$

Here, $[\text{X}]$ denotes molar concentration of X (units: kmol/m^3), $k_G(T)$ is the global rate coefficient, n and m relate to the **reaction order**. According to (4), the reaction (3) is n th order with respect to fuel, m th order with respect to oxidant, and $(m + n)$ th order overall. It is worthwhile to note that m and n are determined from experimental data and are not necessarily integers. The use of global reactions to express chemistry is usually a black box approach and has limited use in combustion. It does not provide a basis for understanding what is actually happening. For ex, let's consider the following global reaction:



It implies that two moles of hydrogen molecule react with one mole of oxygen to form one mole of water, which is not strictly true. In reality many sequential processes occur that involve several intermediate species. Following elementary reactions, among others, are important in conversion of H_2 and O_2 to water:



To have a complete picture of the combustion of H_2 with O_2 , more than 20 elementary reactions can be considered. Reaction mechanism is the collection of elementary reactions to describe the overall reaction. Reaction mechanisms may involve a few steps or as many as several hundred (even thousands).

2 Elementary Reaction Rates

Using the concept of elementary reactions has many advantages. Unlike global reactions which follows a black-box approach and has experimentally determined reaction order, the utility of elementary reactions is showcased in their illustration of complete reaction mechanisms and easily determined reaction order.

Molecularity of the reaction: number of species that form the reaction complex. While reaction order is determined using number of concentration factors in the rate law, molecularity is determined by the number of participating species in a postulated reaction step. The categorisation of elementary reactions based on their molecularity can be done by:

- Unimolecular - involving only one reactant molecule (may include third body molecule for activation)
- Bimolecular - two molecules collide and react to form two different molecules. Most common combustion-related elementary reactions can be represented using this form.
- Trimolecular/ Termolecular - three molecules collide and react to form products. These are usually much rare at normal operating conditions since the probability of collision of 3 molecules at a time is much lower than that of bi-molecular collisions.

2.1 Bimolecular Reactions and Collision Theory

Most combustion related elementary reactions are bimolecular:



The rate at which the reaction proceeds is

$$\frac{d[A]}{dt} = -k_{\text{bimolec}}[A][B] \quad (11)$$

All elementary bimolecular reactions are overall second order, being first order with respect to each of the reacting species. $k_{\text{bimolec}} \propto f(T)$ and has a theoretical basis, unlike k_G , rate coefficient of a global reaction. An estimate of k_{bimolec} can be made using first principle methods and many models exist to compute the reaction rate expression. One such theoretical model is offered by the collision theory which provide insights into the form of rate law as expressed by (11) and to suggest the temperature dependence of the bimolecular rate coefficient. It is still important to note that the collision theory, like many others for bimolecular reactions has several shortcomings. The approach is important for historical reasons and may provide a simple way to visualize bimolecular reactions.

In the study of molecular collisions, one must consider the effect of size of the molecules. For simplicity, let's consider the collision of 2 identical spherical molecules with molecular diameter σ . When these collide, the distance between their centers should be less than or equal to σ . If the distance between their centres is larger than σ , no collision occurs. This is illustrated in Figs.1a and 1b In terms of collision cross section area, the larger the diameter of these molecules, the greater the probability that two molecules will collide. The collision of two molecules each with diameter σ can be considered equivalent to the collision of a molecule with diameter 2σ and another molecule

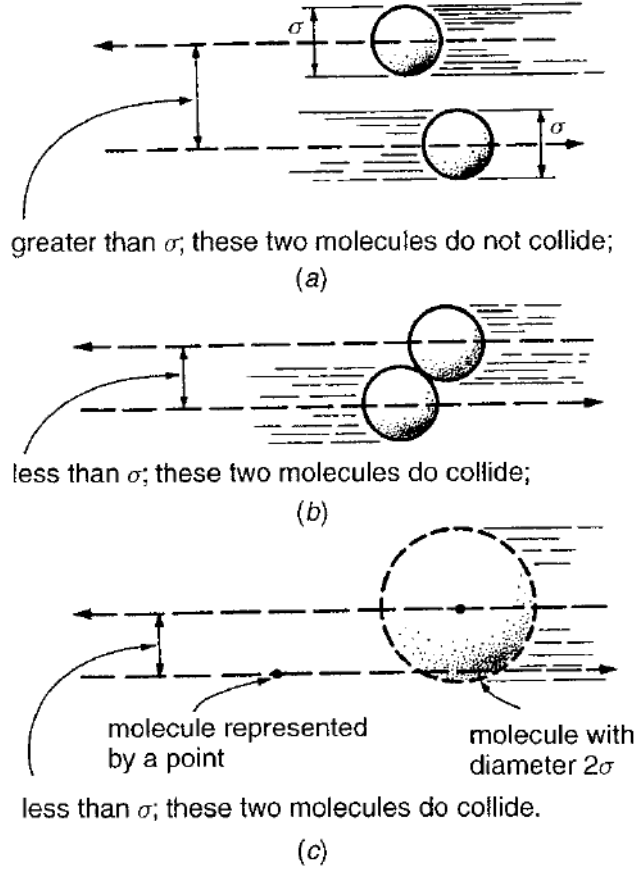


Figure 1: Illustration of the distance required between molecules for a collision to occur

represented as a point mass (Fig. 1c). In each representation, the collision is said to have occurred when the distance between their centers is less than or equal to σ and calculations are simpler with the second representation. This approach is to consider a single molecule of diameter σ travelling at constant speed v and experiencing collisions with identical, but stationary molecules. If the distance travelled (mean free path) between collisions is large, then moving molecule sweeps out a cylindrical volume of $v\pi\sigma^2\Delta t$. For random distribution of stationary molecules with number density n/V , number of collisions

$$Z \equiv \text{collisions per unit time} = (n/V)v\pi\sigma^2 \quad (12)$$

For Maxwellian velocity distribution for all molecules,

$$Z_c = \sqrt{2}(n/V)\pi\sigma^2\bar{v} \quad (13)$$

(13) applies to identical molecules. For different molecules, we can use $\sigma_A + \sigma_B \equiv 2\sigma_{AB}$

$$Z_c = \sqrt{2}(n_B/V)\pi\sigma_{AB}^2\bar{v}_A \quad (14)$$

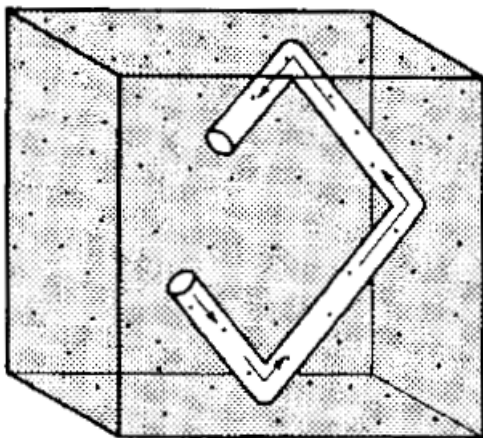


Figure 2: Collisions of a moving molecule of radius σ with stationary point molecules. In the volume swept out by the moving molecule, there are N point molecule masses.

which expresses frequency of collisions of a single moving A molecule with all stationary B molecules (as noted in Fig. 2). For all A molecules colliding with moving B molecules:

$$Z_{AB}/V = (n_A/V)(n_B/V)\pi\sigma_{AB}^2(\bar{v}_A^2 + \bar{v}_B^2)^{1/2} \quad (15)$$

If we express mean molecular speed in terms of temperature,

$$Z_{AB}/V = (n_A/V)(n_B/V)\pi\sigma_{AB}^2\left(\frac{8k_B T}{\pi\mu}\right)^{1/2} \quad (16)$$

where k_B = Boltzmann constant, $\mu = (m_A m_B)/(m_A + m_B)$ = reduced mass, and T = absolute temperature.

We can relate Z_{AB}/V to reaction rates:

$$-\frac{d[A]}{dt} = \left[\begin{array}{c} \text{No. of collisions} \\ \text{A and B molecules} \\ \text{per unit volume} \\ \text{per unit time} \end{array} \right] \cdot \left[\begin{array}{c} \text{Probability that} \\ \text{a collision} \\ \text{leads to} \\ \text{reaction} \end{array} \right] \cdot \left[\frac{\text{kmol of A}}{\text{No. of molecules of A}} \right] \quad (17)$$

or

$$-\frac{d[A]}{dt} = (Z_{AB}/V)\mathcal{P}N_{AV}^{-1} \quad (18)$$

where N_{AV} is the Avogadro number and $\mathcal{P}$ is the probability that a collision leads to reaction. The Swedish chemist and physicist Svante Arrhenius (1859-1927) stated that only those molecules that possess energy greater than a certain amount E_a will react, and these high energy, active molecules lead to products. Since Arrhenius was also the first scientist (1889) to introduce the Boltzmann factor $\exp(-E_a/R_u T)$ to calculate chemical reaction rates. Kinetic theory shows that the Boltzmann factor gives the fraction of all collisions that have a translational kinetic energy along the inter-nuclear axis greater than E_a . The equation

$$k = A \exp\left(\frac{-E_a}{R_u T}\right) \quad (19)$$

is called the Arrhenius law where A is assumed to include the effect of the collision terms, the steric factors associated with the orientation of the colliding molecules, and the mild temperature dependence of the pre-exponential factor. The pre-exponential factor A corresponds to the product of AT^b where the exponent b lies between 0 and 1.

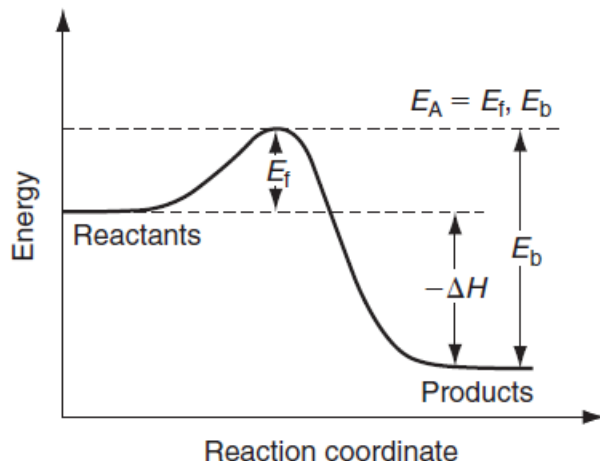


Figure 3: Energy as a function of a reaction coordinate for a reacting system

The energy term in the Boltzmann factor may be considered as the size of the barrier along a potential energy surface for a system of reactants going to products, as shown schematically in Fig.3. The state of the reacting species at this activated energy can be regarded as some intermediate complex that leads to the products. This energy is referred to as the activation energy of the reaction and is generally given the symbol E_a . In Fig.3, this energy is given the symbol E_f , to distinguish it from the condition in which the product species can revert to reactants by a backward reaction. The activation energy of this backward reaction is represented by E_b and is obviously much larger than E_f for the forward step. Fig.3 shows an exothermic condition for reactants going to products. The relationship between the activation energy and the heat of reaction has been developed. Generally, the more exothermic a reaction is, the smaller the activation energy. In complex systems, the energy release from one such reaction can sustain other, endothermic reactions, such as that represented in Fig.3 for products reverting back to reactants. For example, once the reaction is initiated, acetylene will decompose to the elements in a monopropellant rocket in a sustained fashion because the energy release of the decomposition process is greater than the activation energy of the process. In contrast, a calculation of the decomposition of benzene shows the process to be exothermic, but the activation energy of the benzene decomposition process is so large that it will not sustain monopropellant decomposition. For this reason, acetylene is considered an unstable species and benzene a stable one.

Essentially, from (18), the probability ($\mathcal{P}$) that a collision will lead to a reaction can be expressed as a product of two factors:

- an energy factor $\exp[-E_A/(R_u T)]$ which expresses the fraction of collisions that occur with an energy above the activation energy

- a geometrical or steric factor p , that takes into account the geometry of collision.

$$\mathcal{P} = p \cdot \exp \left[\frac{-E_a}{R_u T} \right] \quad (20)$$

With substitutions $n_A/V = [A]N_{AV}$ and $n_B/V = [B]N_{AV}$, (18) becomes:

$$-\frac{d[A]}{dt} = pN_{AV}\sigma_{AB}^2 \left[\frac{8\pi k_B T}{\mu} \right]^{1/2} \cdot \exp \left[\frac{-E_A}{R_u T} \right] [A][B] \quad (21)$$

Comparing (21) with (11)

$$k_{bimolec}(T) = k(T) = pN_{AV}\sigma_{AB}^2 \left[\frac{8\pi k_B T}{\mu} \right]^{1/2} \exp \left[\frac{-E_A}{R_u T} \right] \quad (22)$$

Collision theory is not capable of providing any means to determine E_A or p . More advanced theories do allow calculation of $k(T)$ from first principles to a limited extent. If the temperature range of interest is not too large, k can be expressed by the semi-empirical Arrhenius form

$$k(T) = A \exp \left[\frac{-E_A}{R_u T} \right] \quad (23)$$

where A is a constant termed pre-exponential factor or frequency factor. From (22), we see that A is not strictly constant but depends, based on collision theory, on $T^{1/2}$. Arrhenius plots of $\log k$ versus $1/T$ for experimental data are used to extract values for the activation energy, since the slope of such plots is E_A/R_u . Although tabulation of experimental values for rate coefficients in Arrhenius form is common, current practice frequently utilizes the three-parameter functional form:

$$k(T) = AT^b \exp \left[\frac{-E_A}{R_u T} \right] \quad (24)$$

where A , b , and E_A are three empirical constants.

2.2 Unimolecular Reactions

These are reactions which involves single species undergoing a rearrangement to form one or two product species, i.e.



or



Examples relevant to combustion reactions include dissociation reactions like $O_2 \rightarrow O + O$, $H_2 \rightarrow H + H$, etc. These reactions are typically first order at high pressures

$$\frac{d[A]}{dt} = -k_{uni}[A] \quad (27)$$

At low pressures, the reaction rate may also depend a third molecule that may exist within the reaction volume

$$\frac{d[A]}{dt} = -k[A][M] \quad (28)$$

To explain this interesting behavior requires postulating a three-step mechanism. Since we have yet to explore some of the concepts required to deal with this, we conclude our treatment of unimolecular reactions for the time being. See section 3.5 for detailed explanation.

2.3 Termolecular Reactions

These reactions involves the interaction of 3 molecules and are consequently, quite rare.



Recombination reactions such as $H + H + M \rightarrow H_2 + M$ and $H + OH + M \rightarrow H_2O + M$ are important examples of termolecular reactions in combustion. Termolecular reactions are third order, and their rates can be expressed as

$$\frac{d[A]}{dt} = -k_{\text{ter}}[A][B][M] \quad (30)$$

where, again, M may be any molecule and is frequently referred to as a third body. When A and B are the same species, as in $H + H + M$, a factor of two must multiply the right-hand side of (30) since two A molecules disappear to form C. In radical-radical reactions, the third body is required to carry away the energy liberated in forming the stable species. During the collision, the internal energy of the newly formed molecule is transferred to the third body, M, and is manifest as kinetic energy of M. Without this energy transfer, the newly formed molecule would dissociate back to its constituent atoms.

3 Multi-step Mechanisms

3.1 Net Production Rate

Consider some of the reactions in H_2 - O_2 system



The net production rate of any species, say X, involved is the sum of all of the individual elementary rates producing X minus all of the rates destroying X. Net production rate of O_2 is then,

$$\begin{aligned} \frac{d[O_2]}{dt} = & k_{r1}[HO_2][H] + k_{r2}[OH][O] + k_{r4}[HO_2][M] \\ & - k_{f1}[H_2][O_2] - k_{f2}[H][O_2] - k_{f4}[H][O_2][M] \end{aligned} \quad (32)$$

Net production rate for H atoms:

$$\begin{aligned} \frac{d[H]}{dt} = & k_{f1}[H_2][O_2] + k_{r2}[OH][O] + k_{f3}[OH][H_2] + k_{r4}[HO_2][M] \\ & - k_{r1}[HO_2][H] - k_{f2}[H][O_2] - k_{r3}[H_2O][H] - k_{f4}[H][O_2][M] \end{aligned} \quad (33)$$

For any general species X_i ,

$$\begin{aligned} \frac{d[X_i](t)}{dt} &= f_i\{[X_1](t), [X_2](t), \dots, [X_n](t)\} \\ [X_i](0) &= [X_i]_0 \end{aligned} \quad (34)$$

3.2 Compact Notation

Since mechanisms may involve many elementary steps and many species, a generalized compact notation has been developed for the mechanism and the individual species production rates. For the mechanism,

$$\sum_{j=1}^N \nu'_{ji} X_j \rightleftharpoons \sum_{j=1}^N \nu''_{ji} X_j \quad \text{for } i = 1, 2, \dots, L \quad (35)$$

where ν'_{ji} and ν''_{ji} are stoichiometric coefficients of reactants and products, respectively.

N		L	
j	Species	i	Reaction
1	O ₂	1	(31a)
2	H ₂	2	(31b)
3	H ₂ O	3	(31c)
4	HO ₂	4	(31d)
5	O		
6	H		
7	OH		
8	M		

Stoichiometric coefficient matrices:

$$\nu'_{ji} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \quad (36)$$

$$\nu''_{ji} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (37)$$

Net production rate of each species in a multistep mechanism:

$$d[X_j]/dt \equiv \dot{\omega}_j = \sum_{i=1}^L \nu_{ji} q_i \quad \text{for } j = 1, 2, \dots, N \quad (38)$$

where

$$\nu_{ji} = (\nu''_{ji} - \nu'_{ji}) \quad (39)$$

$$q_i = k_{fi} \prod_{j=1}^N [X_j]^{\nu'_{ji}} - k_{ri} \prod_{j=1}^N [X_j]^{\nu''_{ji}} \quad (40)$$

For example, $q_i (= q_1)$ for reaction (31a) is

$$\begin{aligned} q_i &= k_{f1} [\text{O}_2]^1 [\text{H}_2]^1 [\text{H}_2\text{O}]^0 [\text{HO}_2]^0 [\text{O}]^0 [\text{H}]^0 [\text{OH}]^0 [\text{M}]^0 \\ &\quad - k_{r1} [\text{O}_2]^0 [\text{H}_2]^0 [\text{H}_2\text{O}]^0 [\text{HO}_2]^1 [\text{O}]^0 [\text{H}]^1 [\text{OH}]^0 [\text{M}]^0 \\ &= k_{f1} [\text{O}_2] [\text{H}_2] - k_{r1} [\text{HO}_2] [\text{H}] \end{aligned} \quad (41)$$

Writing similar expressions for $i = 2, 3$, and 4 and summing completes the total rate expression for $\dot{\omega}_j$.

3.3 Rate Coefficients and Equilibrium Constants

At equilibrium forward and reverse reaction rates must be equal.



Formation rate of species A:

$$\frac{d[\text{A}]}{dt} = -k_f [\text{A}] [\text{B}] + k_r [\text{C}] [\text{D}] \quad (43)$$

For equilibrium, time rate of change of [A] must be zero. Same goes for B, C, and D.

$$0 = -k_f [\text{A}] [\text{B}] + k_r [\text{C}] [\text{D}] \quad (44)$$

Rearranging, we get

$$\frac{[\text{C}] [\text{D}]}{[\text{A}] [\text{B}]} = \frac{k_f(T)}{k_r(T)} \quad (45)$$

Previously we have defined equilibrium constant as,

$$K_p = \frac{(p_{\text{C}}/p^o)^c (p_{\text{D}}/p^o)^d \dots}{(p_{\text{A}}/p^o)^a (p_{\text{B}}/p^o)^b \dots} \quad (46)$$

Since molar concentrations are related to mol fractions and partial pressures as,

$$[X_i] = \chi_i p / (R_u T) = p_i / (R_u T) \quad (47)$$

we can define an equilibrium constant based on molar concentrations, K_c and relate it to K_p ,

$$K_p = K_c (R_u T / p^o)^{c+d+\dots-a-b-\dots} \quad (48)$$

or

$$K_p = K_c(R_u T / p^o)^{\Sigma \nu'' - \Sigma \nu'} \quad (49)$$

where, K_c is defined as,

$$K_c = \frac{[C]^c [D]^d \dots}{[A]^a [B]^b \dots} = \frac{\prod_{\text{prod}} [X_i]^{\nu_i''}}{\prod_{\text{react}} [X_i]^{\nu_i'}} \quad (50)$$

So that,

$$\frac{k_f(T)}{k_r(T)} = K_c(T) \quad (51)$$

For bimolecular reactions $K_c = K_p$.

3.4 Steady-State Approximations

Analysis of reactive systems can be simplified by applying steady-state approximation to the reactive species or radicals. Steady-state approximation is justified when the reaction forming the intermediate species is slow, while the reaction destroying the intermediate species is very fast. As a result the concentration of the radical is small in comparison with those of the reactants and products. Example (Zeldovich mechanism for NO formation):



First reaction is slow (rate limiting); while second is fast. Net production rate of N atoms,

$$\frac{d[\text{N}]}{dt} = k_1[\text{O}][\text{N}_2] - k_2[\text{N}][\text{O}_2] \quad (54)$$

After a rapid transient allowing buildup of N, $d[\text{N}]/dt$ approaches zero.

$$0 = k_1[\text{O}][\text{N}_2] - k_2[\text{N}]_{\text{ss}}[\text{O}_2] \quad (55)$$

$$[\text{N}]_{\text{ss}} = \frac{k_1[\text{O}][\text{N}_2]}{k_2[\text{O}_2]} \quad (56)$$

Time rate of change of $[\text{N}]_{\text{ss}}$ is

$$\frac{d[\text{N}]_{\text{ss}}}{dt} = \frac{d}{dt} \left[\frac{k_1[\text{O}][\text{N}_2]}{k_2[\text{O}_2]} \right] \quad (57)$$

3.5 Mechanism for Unimolecular Reactions

Let's consider a three-step mechanism:



In step 1: kinetic energy transferred to A from M; A has increased internal vibrational and rotational energies and becomes an energized A molecule, A^* . Two possible scenarios for A^* :

- A^* may collide with another molecule and goes back to A (58b)
- A^* may decompose into products (58c)

The rate at which products are formed:

$$\frac{d[\text{products}]}{dt} = k_{unim}[A^*] \quad (59)$$

Net production rate of A^* :

$$\frac{d[A^*]}{dt} = k_e[A][M] - k_{de}[A^*][M] - k_{unim}[A^*] \quad (60)$$

Steady-state approximation for A^* , i.e. $d[A^*]/dt = 0$,

$$[A^*] = \frac{k_e[A][M]}{k_{de}[M] + k_{unim}} \quad (61)$$

Substitute (61) into (59),

$$\frac{d[\text{products}]}{dt} = \frac{k_e[A][M]}{(k_{de}/k_{unim})[M] + 1} \quad (62)$$

Another form of writing the overall rate of production of products for the overall reaction:



$$-\frac{d[A]}{dt} = \frac{d[\text{products}]}{dt} = k_{app}[A] \quad (64)$$

k_{app} is an apparent unimolecular rate coef. Equating (62) and (64) yields

$$k_{app} = \frac{k_e[M]}{(k_{de}/k_{unim})[M] + 1} \quad (65)$$

(65) lets us to explain pressure dependence of unimolecular reactions.

- At high enough pressures ($k_{de}[M]/k_{unim}) \gg 1$ because $[M]$ increases as the pressure is increased; then

$$k_{app}(P \rightarrow \infty) = k_{unim}k_e/k_{de} \quad (66)$$

- At low enough pressures ($k_{de}[M]/k_{unim}) \ll 1$, then

$$k_{app}(P \rightarrow 0) = k_e[M] \quad (67)$$

3.6 Chain Breaking and Chain Branching Reactions

Chain reactions produce one or more radical species that subsequently react to produce another radical(s). To learn some of the features of chain reactions, we consider a hypothetical chain mechanism, which is globally represented as



Chain initiation:



Chain-propagating reactions involving free radicals A and B:



Chain-terminating reaction is:



where AB is the stable product. Concentrations of A and B are small throughout the course of reaction, so is AB at initial stages; thus we ignore reverse reactions and determine reaction rates for stable species as:

$$\frac{d[A_2]}{dt} = -k_1[A_2][M] - k_3[A_2][B] \quad (68)$$

$$\frac{d[B_2]}{dt} = -k_2[B_2][A] \quad (69)$$

$$\frac{d[AB]}{dt} = k_2[A][B_2] + k_3[B][A_2] + k_4[A][B][M] \quad (70)$$

If we use the steady-state approximation for free radicals A and B:

$$\frac{d[A]}{dt} = 2k_1[A_2][M] - k_2[A][B_2] + k_3[B][A_2] - k_4[A][B][M] = 0 \quad (71)$$

$$\frac{d[B]}{dt} = k_2[A][B_2] - k_3[B][A_2] - k_4[A][B][M] = 0 \quad (72)$$

Simultaneous solution of (71) and (72) for A:

$$[A] = \frac{k_1}{2k_2} \frac{[M][A_2]}{[B_2]} + \frac{k_3}{k_4} \frac{[A_2]}{[M]} \cdot \left\{ \left[1 + \left(\frac{k_1 k_4}{2k_2 k_3} \frac{[M]^2}{[B_2]} \right)^2 \right]^{1/2} - 1 \right\} \quad (73)$$

A similar expression can be written for [B]. With the steady-state values for [A] and [B], $d[A_2]/dt$, $d[B_2]/dt$, and $d[AB]/dt$ can be found for initial values of $[A_2]$ and $[B_2]$. If we consider the simplest of the three, $d[B_2]/dt$:

$$\frac{d[B_2]}{dt} = -\frac{k_1}{2}[A_2][M] - \frac{k_2 k_3}{k_4} \frac{[A_2][B_2]}{[M]} \left\{ \left[1 + \left(\frac{k_1 k_4}{2k_2 k_3} \frac{[M]^2}{[B_2]} \right)^2 \right]^{1/2} - 1 \right\} \quad (74)$$

The last two Eqns can be further simplified:

$$k_1 k_4 [M]^2 / (2k_2 k_3 [B_2]) \ll 1 \quad (75)$$

since k_2 and k_3 must be much larger than k_1 and k_4 for steady-state approximation to apply. (73) and (74) can be approximated as:

$$[A] \approx \frac{k_1}{2k_2} \frac{[M][A_2]}{[B_2]} + \frac{k_1^2 k_4}{8k_2^2 k_3} \frac{[M]^3 [A_2]}{[B_2]^2} \quad (76)$$

$$\frac{d[B_2]}{dt} \approx -\frac{k_1}{2} [A_2][M] - \frac{k_1^2 k_4}{4k_2 k_3} \frac{[M]^3 [A_2]}{[B_2]} \quad (77)$$

First term in both (76) and (77) dominates at low pressures. Concentration of A depends on the ratio k_1/k_2 . Rate at which B_2 disappears is governed by k_1 . Increasing k_2 and k_3 increases radical concentration, but has no effect on production rate of products. $[A]$ and $[B]$ are directly proportional to p . Reaction rates of major species scale with pressure squared provided that second terms in (76) and (77) remain small. At high pressures, the second terms in (76) and (77) become important (they increase with pressure faster than the first terms do). So, k_4 has some influence at high pressures, although it does not have much influence on radical concentrations or overall reaction rate at lower pressures.

Chain-branching reactions involve the formation of two radical species from a reaction that consumes only one radical.



4 Chemical Time Scales

Magnitude of chemical times relative to convective or mixing times is of importance in combustion.

4.1 Unimolecular reactions



$$-\frac{d[A]}{dt} = \frac{\text{products}}{dt} = k_{app}[A] \quad (80)$$

Integrating at constant T for $[A](t=0)=[A]_0$,

$$[A](t) = [A]_0 \exp[-k_{app}t] \quad (81)$$

Characteristic chemical time can be defined as the time required for the concentration of A to fall from its initial value to a value equal to $1/e$ times the initial value,

$$\frac{[A](\tau_{chem})}{[A]_0} = 1/e \quad (82)$$

Combining (81) and (82) yields,

$$1/e = \exp[-k_{app}\tau_{chem}] \quad (83)$$

or

$$\tau_{chem} = 1/k_{app} \quad (84)$$

4.2 Bimolecular Reactions



$$\frac{d[A]}{dt} = -k_{\text{bimolec}}[A][B] \quad (86)$$

For this single reaction $[A]$ and $[B]$ are related by stoichiometry. Any change in $[A]$ has a corresponding change in $[B]$,

$$x \equiv [A]_0 - [A] = [B]_0 - [B] \quad (87)$$

$$[B] = [A] + [B]_0 - [A]_0 \quad (88)$$

Substituting (87) into (11) and integrating,

$$\frac{[A](t)}{[B](t)} = \frac{[A]_0}{[B]_0} \exp \{([A]_0 - [B]_0)k_{\text{bimolec}}t\} \quad (89)$$

Substitute (88) into (89), and set:

$$[A]/[A]_0 = 1/e \quad \text{when} \quad t = \tau_{\text{chem}} \quad (90)$$

gives

$$\tau_{\text{chem}} = \frac{\ln [e + (1 - e)([A]_0/[B]_0)]}{([B]_0 - [A]_0)k_{\text{bimolec}}} \quad (91)$$

If $[B]_0 \gg [A]_0$,

$$\tau_{\text{chem}} = \frac{1}{[B]_0 k_{\text{bimolec}}} \quad (92)$$

4.3 Termolecular Reactions



For a simple system at constant T , $[M]$ is constant

$$\frac{d[A]}{dt} = (-k_{\text{ter}}[M])[A][B] \quad (94)$$

where $(-k_{\text{ter}}[M])$ plays the same role as k_{bimolec} does for a bimolecular reaction. Then, the characteristic time for termolecular reactions is,

$$\tau_{\text{chem}} = \frac{\ln [e + (1 - e)([A]_0/[B]_0)]}{([B]_0 - [A]_0)k_{\text{ter}}[M]} \quad (95)$$

And, when $[B]_0 \gg [A]_0$,

$$\tau_{\text{chem}} = \frac{1}{[B]_0(k_{\text{ter}}[M])} \quad (96)$$